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ORIGINAL ARTICLE

First successful domestication and determination of nutritional and antioxidant properties of the red ear mushroom *Auricularia thailandica* (*Auriculariales, Basidiomycota*)

Asanka R. Bandara[1,2,3] · **Samantha C. Karunarathna**[1,4] · **Peter E. Mortimer**[1,4] · **Kevin D. Hyde**[1,2,3,4] · **Sehroon Khan**[1,4] · **Pattana Kakumyan**[2,3] · **Jianchu Xu**[1,4]

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Abstract In order to fulfill the increasing demand for edible mushrooms and maintain a steady supply throughout the year, it is necessary to domesticate new wild edible mushroom species. In this study, we domesticated *Auricularia thailandica*, a newly identified edible species that was collected from the wild in Thailand. We compared the growth of *A. thailandica* in three different growth media: sawdust, wheat husk and sugarcane bagasse. The use of sawdust resulted in the fastest rate of mycelial colonisation at 56.4 ± 1.2 days, pinheads were formed in 14.2 ± 0.4 days and the biological efficiency was 17.1 ± 2.8%. Nutritional analysis showed that *A. thailandica* contained a higher proportion of protein (12.99 ± 0.05%) and fat (2.93 ± 0.66%) by dry weight than other commercial *Auricularia* spp. available on the market. The antioxidant potential of *A. thailandica* varied with the extraction solvent used, and the effective concentration was found to be significantly lower in methanol extract than aqueous extract. Additionally, this species has an attractive reddish orange colour and larger basidiocarps than other commercially available species.

Keywords Antioxidant · Edible mushrooms · Southeast Asia · Thailand

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✉ Kevin D. Hyde
kdhyde3@gmail.com

✉ Jianchu Xu
jxu@mail.kib.ac.cn

[1] Key Laboratory for Plant Diversity and Biogeography of East Asia, Kunming Institute of Botany, Chinese Academy of Sciences, Kunming 650201, Yunnan, China

[2] Center of Excellence in Fungal Research, Mae Fah Luang University, Chiang Rai 57100, Thailand

[3] School of Science, Mae Fah Luang University, Chiang Rai 57100, Thailand

[4] World Agroforestry Centre, East and Central Asia, Kunming 650201, Yunnan, China

Introduction

Auricularia Bull. species have a long history of being used as an edible and medicinal mushroom in Chinese and European folk cultures (Sękara et al. 2015). Numerous studies have reported *Auricularia* to have many pharmacological properties, including antitumour, anticoagulant, antioxidant and cholesterol-lowering effects (Chen et al. 2011; Cheung 1996; Elisashvili 2012; Sękara et al. 2015). Additionally, *Auricularia* has been shown to have anti-inflammatory (Damte et al. 2011; Ukai et al. 1983), antimicrobial (Gbolagade and Fasidi 2005) and immunomodulatory properties (De Silva et al. 2012a; Sheu et al. 2004). *Auricularia* species are higher in fibre than other commercially available edible mushrooms (Cheung 1997; Misaki and Kakuta 1995). As a result of this high fibre content, researchers have shown that the use of *Auricularia* significantly improved constipation-related symptoms without any side effects in patients with functional constipation in vivo (Kim et al. 2004). Compared to the common edible jelly mushroom (*Tremella fuciformis*), *Auricularia* contains higher amounts of crude proteins and lower amounts of carbohydrates (Cheung 2013; Mau et al. 1998). Edible *Auricularia* contain very low levels of soluble sugars, which causes the absence of sweet perception (Mau et al. 1998) and showed hypoglycaemic effects in vivo (De Silva et al. 2012b; Yuan et al. 1998). Unlike other edible mushrooms, *Auricularia* has a low level of fat content (Cheung 2013) and 60% of the fatty acids contained in wild *Auricularia* are unsaturated (Kavishree et al. 2008). Consuming *Auricularia* could be

helpful for patients with high blood cholesterol; as such, patients are often advised to adopt low-fat diets and a higher proportion of unsaturated rather than saturated fatty acids (Lichtenstein et al. 2006). In addition, *Auricularia* have the potential to prevent hypolipidaemic activity, which is a contributory factor to obesity and associated health problems (Reza et al. 2015). *Auricularia* is highly regarded for its medicinal and nutritional benefits in East Asia, and is consequently highly popular and extensively cultivated in the region (Buot 1994; Chandrasrikul 1996; Chang and Lee 2004; Duc 2005; Irawati et al. 2012; Peng 2008; Razak et al. 2013; Reyes et al. 2009; Tapingkae 2005; Wu et al. 2014).

Auricularia is commonly produced on sawdust substrates with various supplements in plastic bags, as well as on logs of various trees (Chang and Miles 2004). In addition, maize cobs, wheat straw, grass straw, sugarcane bagasse (Onyango et al. 2011) and coffee pulp (Martínez-Carrera et al. 2000) have been used as a main substrate, while rice bran, wheat bran (Onyango et al. 2011; Sales-Campos et al. 2011) and oil palm wastes (Razak et al. 2013) have been used as supplements. China is the world's top producer of *Auricularia* (Zhang et al. 2014) and annual production in 2011 reached nearly 4.9 million tonnes (Wu et al. 2013). *Auricularia* is the fourth most cultivated mushroom genus after *Agaricus*, *Lentinula* and *Pleurotus* in the world (Chang and Miles 2004; Tang et al. 2010). Four species [*Auricularia auricula-judae* (Bull.: Fr.) Quél. (= *A. auricula* (L. ex Hook.) Underw.), *A. fuscosuccinea* (Mont.) Henn., *A. heimuer* F. Wu et al. and *A. polytricha* (Mont.) Sacc.] are commercially cultivated throughout the world (Chang and Miles 2004; Morales et al. 2000; Sękara et al. 2015; Wu et al. 2014).

Recently, several new *Auricularia* species were introduced from different parts of the world (Bandara et al. 2015, 2017; Looney et al. 2013; Wu et al. 2014, 2015a, b). Among them, preliminary research has shown that *A. thailandica* Bandara & K.D. Hyde may have potential for domestication for edible use in tropical and subtropical climates. This species is widely distributed in Southeast Asia (Bandara et al. 2015) and wild specimens are frequently harvested by local people for their own consumption and sale (Fig. 1).

Wild *Auricularia* can only be collected from Southeast Asian forests during the rainy season (May to September). The domestication of new edible species of *Auricularia* could provide nutritional benefits and livelihood opportunities for farmers in the region. We, therefore, domesticated a newly identified *Auricularia* species, *A. thailandica*, compared its growth in different substrates and analysed the nutritional and antioxidant properties of this species.

Materials and methods

Isolation and spawn production

A pure mycelium culture was established from the wild *A. thailandica* strain MFLU162122 (AB102) using tissue culture. The mycelia were subcultured on potato dextrose agar (Oxoid, CM0139) at 28 °C for 10 days and transferred to cooked rye grain and incubated at 28 °C for 2 weeks.

Mycelia growth and basidiocarp production

Oak (*Quercus* sp.) sawdust, ground sugarcane bagasse and ground wheat husk (particle size ≤ 3 mm) were separately mixed with rice bran in a 79:20 ratio. Calcium sulphate (1%) was added to each mixture and mixed with water until the moisture level reached about 65%. The substrate mixtures were packed (1 kg per bag) tightly in plastic bags and autoclaved at 121 °C and 15 psi for 4 h. The bags were allowed to cool for 24 h and then inoculated with 1% (w/w) of mycelia colonised rye grains. Inoculated bags were incubated at 28 °C in the dark. The time required to complete colonisation of mycelia in 30 bags of each substrate was recorded every 4 days. When the substrate was fully invaded by mycelia, bags were cut and opened at the top. The temperature for the fruiting phase was 28 °C and the humidity was 85–95%. Mushroom yield was recorded for each flush for 30 bags and the biological efficiency was calculated per dry weight of substrate. The calculation used to determine the biological

Fig. 1 Basidiocarps of *Auricularia thailandica*: **a** on a decaying trunk in the wild, **b** fresh and **c** dry, at the roadside mushroom selling huts at Doi Mae Salong in Northern Thailand

efficiency was: Biological efficiency (%) = Yield of fresh basidiocarp (g)/substrate dry weight (g) × 100 (Liang et al. 2016). Basidiocarps of *A. thailandica* produced on sawdust medium were used to quantify the nutrient content, mineral content and antioxidant activity.

Quantification of nutrient content

Proximate analysis was carried out according to the standard procedure of the Association of Official Analytical Chemists (AOAC 1995). The moisture content was measured by drying 2 g of fresh sample at 105 °C ± 5 °C in a hot air oven (DHG9070A, Shanghai Yiheng Technical Co. Ltd.). The ash content was determined by incinerating 1 g of dry sample at 550 °C in a muffle furnace (SX2-25-1, Nanjing Electric Furnace Factory Co. Ltd.). The crude fibre content was measured by treating 2 g of dry sample with 1.25% (w/v) H_2SO_4 and 1.25% (w/v) NaOH, followed by subsequent heating at 105 °C ± 5 °C in a hot air oven (DHG9070A, Shanghai Yiheng Technical Co. Ltd.), until a constant weight was obtained. The Kjeldahl method was used to analyse the organic nitrogen content and the protein content was calculated by multiplying by the conversion factor 6.25 (AOAC 1995). The fat content was determined by continuous extraction of approximately 0.5 g of pre-dried sample with petroleum ether in Soxhlet Extraction (SZF-06A, Shanghai Xinjia Electronic Co., Ltd., Shanghai, China) (AOAC 1995). Total soluble sugars were analysed by using the phenol-sulphuric acid method (Nielsen 2010).

Total soluble free amino acids were determined by the Ninhydrin colourimetric method (Chen et al. 2009). The amino acid composition was analysed by hydrolysing the dried sample with 6 mol/L on a Biochrom 20 Amino Acids Analyser (Biochrom Ltd., Cambridge, UK), in accordance with the methods of Chinese Standard GB/T 5009.124-2003 (Gong et al. 2013).

Quantification of mineral content

Minerals were analysed based on an atomic absorption spectrophotometric (AAS) technique. Dry samples were powdered and 1 g digested with HNO_3 and H_2O_2 at low temperature for 2 h. The concentrations of Cd, Cr, Cu, Fe, Mg, Mn, Ni, Pb and Zn were determined via graphite furnace atomic absorption spectrometry with a novAA® 400P (Analytik Jena, Germany) spectrometer. The concentrations of As and Hg were determined via atomic fluorescence spectroscopy with a PF5 series atomic fluorescence spectrophotometer (Beijing Pu Analysis of General Equipment LLC). For calibration of the AAS technique, different concentrations of standard solutions of each element were prepared from standard stock (1000 μg/mL, National Nonferrous Metals and Electronic Materials Analysis and Testing Center, Beijing, China) by dilution. The detection limit values of elements in the AAS method were 0.082 mg/L for Fe, 0.018 mg/L for Zn, 0.033 mg/L for Mn, 0.015 mg/L for Cu, 0.086 mg/L for Mg, 0.091 mg/L for Ni, 0.075 mg/L for Cr, 0.052 μg/L for Pb, 0.035 μg/L for Cd, 0.0076 μg/L for As and 0.0049 μg/L for Hg.

Preparation of aqueous and methanol extracts

Cultivated, hot air dried basidiocarps were blended, soaked in 100% methanol at 1:3 (w/v) and extracted three times using an ultrasonic bath (VGT2120QTD, Shenzhen Very Good Cleaning Equipment Co. Ltd.) at 45 °C for 40 min. The aliquots were filtered through filter paper Shuangquan No. 201 and evaporated using a vacuum rotary evaporator (Heidolph Instruments GmbH & Co.KG) at 45 °C and 0.075 Mpa. Aqueous extracts were prepared by boiling hot air dried samples (100 g/L) at 100 °C three times, each for 30 min. The aliquots were filtered through Shuangquan No. 201 filter paper and the filtrate was freeze-dried.

Total phenolic content analysis

The total phenolic content of the aqueous and methanolic extracts was estimated using the Folin–Ciocalteu assay (Ainsworth and Gillespie 2007), with minor modifications. One millilitre (1 mL) of extract was thoroughly mixed with 5 mL of Folin–Ciocalteu reagent and the mixture was left at room temperature for 10 min. Then, 4 mL of 7.5% sodium carbonate was added to the mixture and further incubated for 60 min. Then, the absorbance was measured at 765 nm using a spectrophotometer (WFJ7200, UNICO Shanghai Instrument Co., Ltd.). Gallic acid (0.05 mg/mL) was used as a standard and the total phenolic content was expressed in μg of gallic acid equivalents (GAE) for mg of extract.

Reducing power assay

The reducing power was estimated by a ferricyanide/Prussian blue assay (Berker et al. 2010) with adjustment of volume in the final step, i.e. 1 mL was withdrawn from an aliquot of ferricyanide/trichloroacetic acid mixture and the final volume was set to 4 mL with ferric chloride and water. The absorbance was measured at 700 nm using a spectrophotometer (WFJ7200, UNICO Shanghai Instrument Co., Ltd.). Ascorbic acid was used as a standard.

2,2-Diphenyl-1-picrylhydrazyl (DPPH) radical scavenging activity assay

Two millilitres (2 mL) of 0.05 mM methanol DPPH solution (Tianjin Fengchuan Chemical Reagent Technology Co., Ltd.) was added to 1 mL of extract at different concentrations (1–5 mg/mL) and the mixture was incubated for 30 min in the

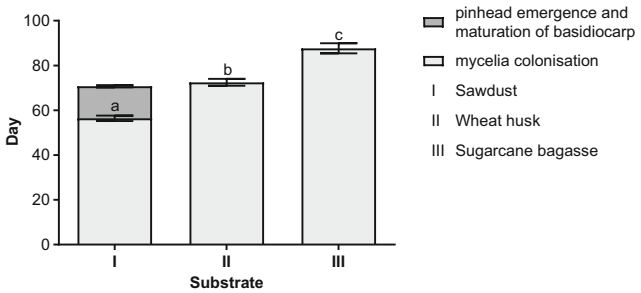

Fig. 2 Time required for complete colonisation of 1 kg of sawdust, wheat husk and sugarcane bagasse substrates by *Auricularia thailandica*. Letters *a–c* denote the mean durations for complete colonisation of three substrates which are significantly different at $p < 0.05$ by Tukey's test

dark. The mixture was centrifuged for 15 min and the change in absorbance of supernatant was measured at 517 nm using a spectrophotometer (WFJ7200, UNICO Shanghai Instrument Co., Ltd.). Ascorbic acid was used as a standard and radical scavenging activities were expressed as a percentage (%) of DPPH quenched. EC50 was determined using GraphPad Prism software version 7.0.

2,2′-Azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) radical scavenging activity assay

The assay was carried out according to the improved ABTS de-colourisation assay (Re et al. 1999). ABTS radicals were produced by mixing 16.5 mL of 2.8 mM ABTS solution with 16.5 mL of 2.44 mM potassium persulphate and were kept in the dark at room temperature for 12–16 h. The solution was diluted with absolute ethanol to give an absorbance of 0.7 ± 0.05 at 734 nm using a spectrophotometer (WFJ7200, UNICO Shanghai Instrument Co., Ltd.). Then, a 100 μL extract at different concentrations (1–5 mg/mL) were added to 3.8 mL of diluted solution and absorbance was measured at 734 nm after 6 min of incubation at room temperature. Ascorbic acid was used as a standard and radical scavenging activities were expressed as the percentage (%) of ABTS quenched. EC50 was determined using GraphPad Prism software version 7.0.

Statistical analysis

Thirty bags of each substrate (three substrates, see Fig. 2) were used to determine the mycelia growth and basidiocarp production, and three replicates of *A. thailandica* samples were used for all the other analyses. The results were analysed using GraphPad Prism statistical software (version 7.0). An unpaired *t*-test was used to determine significant differences between the total phenolic content of the aqueous and methanolic extracts at $p < 0.05$. Analysis of variance (ANOVA) was performed using one-way ANOVA and significant differences ($p < 0.05$) between means were determined using Tukey's multiple range test.

Results

Mycelia growth and basidiocarp production

Mycelia of *A. thailandica* were grown on all substrates (sawdust, sugarcane bagasse, wheat husk) and the time required to fully colonise 1 kg of each substrate was measured (Fig. 2). It took 56.4 ± 1.2 days to colonise 1 kg of sawdust medium, which recorded the highest ($p < 0.05$) growth rate. The lowest ($p < 0.05$) rate of growth was observed on sugarcane bagasse medium, which required 87.6 ± 2.2 days to colonise the same weight (Fig. 2).

Sawdust was used as the substrate for basidiocarp production and preliminary domestication of *A. thailandica*, since it supported the highest mycelia growth rate. The time duration from pinhead emergence until maturation of basidiocarps in the first harvest was 14.2 ± 0.4 days (Fig. 2). The average yield of the first three flushes on the sawdust medium was 60.8 ± 8.8 g and the biological efficiency was 17.1 ± 2.8% (Fig. 3).

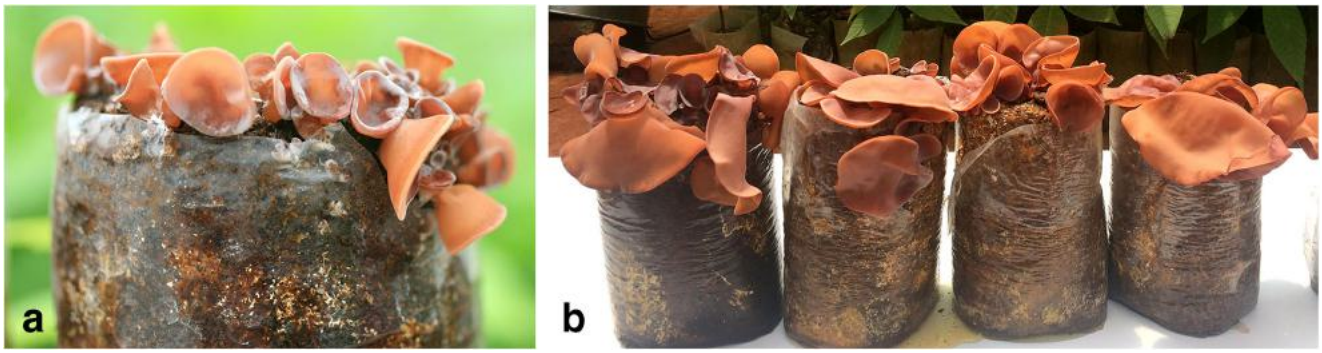

Fig. 3 Cultivated basidiocarps of *Auricularia thailandica* on sawdust medium: **a** early stage and **b** mature stage of basidiocarps

Table 1 Proximate composition, total soluble amino acids and total soluble sugar content of *Auricularia thailandica* (% of dry weight)

Nutrient constituent	Quantity (%)
Moisture	80.75 ± 0.20[a]
Ash	4.30 ± 0.02
Fat	2.93 ± 0.66
Fibre	4.62 ± 0.35
Protein	12.99 ± 0.05
Total soluble free amino acids	1.13 ± 0.01
Total soluble sugar	16.23 ± 0.13

[a] Represented as % of fresh weight

Quantification of nutrient content

The proximate composition, total soluble free amino acids and total soluble sugar content of *A. thailandica* are given in Table 1.

Amino acid analysis showed that 17 amino acids were present in *A. thailandica* (Table 2). The amino acid found at the highest concentration was valine (0.97%), whereas methionine (0.011%) had the lowest concentration. Eight of these 17 amino acids were essential amino acids, meaning that they must be consumed and cannot be synthesised by the human body. Nine were non-essential amino acids. The total essential amino acid content (4.24%) was higher than the total non-essential amino acid content (3.76%).

Quantification of mineral content

The quantities of 15 minerals in *A. thailandica* are given in Table 3. K is the most abundant macroelement and quantities were in descending order at $p < 0.05$: K > P > Mg ≥ Ca > Na. Fe is the most abundant microelement and quantities of microelements were in descending order at $p < 0.05$: Fe > Zn > Mn ≥ Ni ≥ Cu ≥ Cr. Macroelements were found at significantly higher ($p < 0.05$) concentrations than microelements. Toxic elements (As, Cd, Hg and Pb) were contained at significantly lower ($p < 0.05$) concentrations than macro- and microelements in *A. thailandica* and are in tolerable and safe intake levels for humans according to the Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO) and the European Union Commission (EC) regulation no. 1881/2006 (Table 7).

Antioxidant activity

The mean values of ABTS and DPPH free radical scavenging activities upon the concentration gradient of aqueous and methanolic extracts of *A. thailandica* are shown in Fig. 4. The total phenolic content and antioxidant activity of aqueous and methanol extracts of *A. thailandica* are given in Table 4.

Discussion

Auricularia has been reported as producing extracellular enzymes that may aid in the degradation of lignocellulose-based substrates (Adenipekun et al. 2015; Lu and Tang 1986; Ma et al. 2011). Cellulose and lignin are essential to mycelia growth and basidiocarp production of *Auricularia* (Adenipekun et al. 2015; Irawati et al. 2012). The duration of substrate colonisation by *A. thailandica*, therefore, depends on the cellulose and lignin content of the substrate. As shown in Table 5, the higher cellulose and lignin content in sawdust may have enhanced the mycelial growth rate of *A. thailandica*. Moreover, previous studies have shown that mycelia growth in substrates with high nitrogen content is significantly slower than in low-nitrogen substrates (Razak 2013; Yang et al. 2013). Our results showed that the mycelia growth of *A. thailandica* was most efficient on sawdust containing relatively low amounts of nitrogen (Table 5). The biological efficiency of *A. thailandica* (17.1 ± 2.8%) was not significantly different from the biological efficiency of *A. polytricha*, which was previously reported (Liang et al. 2016; Razak et al. 2013). This lack of difference in biological efficiency was likely due to the fact that *A. polytricha* was

Table 2 Amino acid composition of *Auricularia thailandica* (% of dry weight)

Amino acid	Quantity (%)	Amino acid	Quantity (%)
ALA alanine	0.530 ± 0.005	LYS lysine[a]	0.581 ± 0.003
ARG arginine	0.630 ± 0.002	MET methionine[a]	0.011 ± 0
ASP aspartic acid	0.351 ± 0.004	PHE phenylalanine[a]	0.440 ± 0.004
CYS cystine	0.152 ± 0.001	PRO proline	0.362 ± 0.002
GLU glutamic acid	0.921 ± 0.008	THR threonine[a]	0.584 ± 0.004
GLY glycine	0.292 ± 0.001	TYR tyrosine	0.211 ± 0.001
HIS histidine[a]	0.522 ± 0.005	SER serine	0.312 ± 0.006
ILE isoleucine[a]	0.400 ± 0.002	Val valine[a]	0.971 ± 0.009
LEU leucine[a]	0.730 ± 0.007	Total amino acids	8.004

[a] Essential amino acid

Table 3 Mineral content of *Auricularia thailandica* (mg/kg of dry weight)

Macroelement	Quantity (mg/kg)	Microelement	Quantity (mg/kg)	Toxic element	Quantity (mg/kg)
Ca	885.19 ± 2.78	Cr	4.63 ± 0.17	As	0.0690 ± 0.0037
K	13,780.4 ± 8.62	Cu	6.82 ± 0.12	Cd	0.0672 ± 0.0012
Mg	895.74 ± 6.95	Fe	64.38 ± 0.70	Hg	0.0005 ± 0
Na	113.99 ± 1.59	Mn	8.72 ± 0.06	Pb	0.0536 ± 0.0041
P	3880.3 ± 6.95	Ni	7.33 ± 1.13		
		Zn	49.37 ± 0.25		

cultivated on a sawdust medium mixed with a similar ratio of supplements (10–30%) (Liang et al. 2016; Razak et al. 2013) as that used in our study.

Mushrooms are highly valued as a good source of protein, which is an essential part of the human diet (Ingram 2002). The newly domesticated *A. thailandica* mushroom contains higher levels of protein than the commercially cultivated *Auricularia* species (Table 6) (Crisan and Sands 1978; Kadnikova et al. 2015; USDA 2015). Proteins are biopolymers of amino acids. Our analysis showed that *A. thailandica* provides eight out of nine essential amino acids, as well as arginine, cysteine and tyrosine, which are considered semi-essential amino acids and are important for paediatric nutrition. A similar composition with different quantities of amino acids has been observed in wild-collected *A. auricula-judae* (Afiukwa et al. 2015).

Our results indicate that *A. thailandica* is higher in amino acids than *A. polytricha* and both varieties of *A. fuscosuccinea* (Table 6) (Mau et al. 1998), although *A. polytricha* has a higher quantity (17.9%) of soluble sugars (Mau et al. 1998) than *A. thailandica*. Soluble free amino acids and soluble sugars are non-volatile compounds which add flavour to edible mushrooms (Litchfield 1967). However, *A. thailandica* is low in soluble free amino acids and soluble sugars compared to other common edible mushrooms (Kim et al. 2009), which may explain the mild taste of *A. thailandica*.

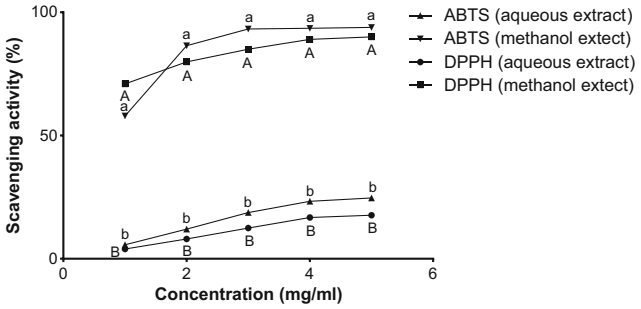

Fig. 4 Free radical scavenging effect of the aqueous and methanol extracts taken from *Auricularia thailandica*. Standard error of the mean < 0.2, $n = 3$. Letters *a*, *b* and *A*, *B* refer to significant differences between mean values at $p < 0.05$ of ABTS and DPPH scavenging activity, respectively. *ABTS* 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) radical scavenging activity, *DPPH* 2,2-diphenyl-1-picrylhydrazyl radical scavenging activity

Maintaining a high intake of dietary fibre has multiple health benefits, including helping in weight regulation and prevention of diseases, such as cardiovascular disease, diabetes and cancer (Cheung 2013). Some studies have reported on the crude fibre content of *Auricularia* species, but these estimates have varied widely. *Auricularia auricula-judae* was reported to have more than 50% higher fibre content than other edible mushrooms (Cheung 1997; Sękara et al. 2015). Yet, other studies found the same species to have less than 10% of the fibre content of other edible mushrooms (Aletor 1995; Crisan and Sands 1978). The crude fibre contents of *A. fuscosuccinea* and *A. polytricha* have been recorded as 11.7% and 3.6% of dry weight, respectively (Mau et al. 1998). In our study, the crude fibre content of *A. thailandica* was calculated as 4.62 ± 0.35% of dry weight.

Auricularia thailandica contains higher levels of crude fat (2.93 ± 0.66%) than *A. auricula-judae* and *A. polytricha*, but lower than *A. fuscosuccinea* (Cheung 1997; Crisan and Sands 1978; Kadnikova et al. 2015; Mau et al. 1998; USDA 2015). According to the fatty acid analysis, 60.2% of the fatty acids contained in *A. polytricha* are unsaturated (Kavishree et al. 2008). Unsaturated fatty acids and especially n-3 polyunsaturated fatty acids may reduce the risk of cardiovascular diseases (Helfant et al. 1970). The crude ash content of commercialised *Auricularia* varies from 0.15% (USDA 2015) to 5.5% (Mau et al. 1998). *Auricularia thailandica* contained 4.30 ± 0.02%

Table 4 Total phenolic content and antioxidant activity of aqueous and methanol extracts of *Auricularia thailandica*. Superscripts [a], [b] and [c], [d] refer to significant differences between mean values at $p < 0.05$ using an unpaired *t*-test and Tukey's test, respectively

Extracts	Total phenolic content (equivalent of μg gallic acid/mg extracts)	DPPH radical scavenging properties EC50 (μg/mL)	ABTS radical scavenging properties EC50 (μg/mL)
Methanol extract	1.99 ± 0.01[a]	341.9 ± 0.8[c]	834.9 ± 1.5[c]
Aqueous extract	0.68 ± 0.01[b]	21,598.3 ± 23.0[d]	14,196 ± 33.5[d]

Ascorbic acid was used as a standard and significant differences in Tukey's test ($p < 0.05$) were observed with both methanol and aqueous extracts for DPPH and ABTS radical scavenging properties

Table 5 Cellulose, lignin and nitrogen content (as percentage values) of the media used in this study

Quantity (%)	Sawdust	Wheat husk	Sugarcane bagasse	References
Cellulose	42–44	36	35	Bledzki et al. (2010); Bodirlau et al. (2007); Rezende et al. (2011)
Lignin	25–31	16	16–24	Bledzki et al. (2010); Bodirlau et al. (2007); Masarin et al. (2011)
Nitrogen	0.86	–	1.2	Hoa et al. (2015)

of crude ash based on its dry weight. However, the level of ash content can only be used to give a rough approximation of the mineral content of basidiocarps (Afiukwa et al. 2015; Wang et al. 2014).

According to the mineral content analysis (Table 3), potassium (K) is the most prevalent element in *A. thailandica*. Previous studies have reported that this is the case for most wild growing and cultivated mushrooms (20,000–40,000 mg/kg) (Kalač 2010, 2013; Wang et al. 2014), including *Auricularia* spp. such as *A. delicata* (3.84%) and *A. polytricha* (2.16%) (Hassan et al. 2017). Wild and cultivated edible mushrooms have also been found to have high levels of P; for example, *Auricularia* spp. were found to have 5000–10,000 mg/kg (Hassan et al. 2017; Kalač 2010, 2013; Wang et al. 2014). *Auricularia delicata*, *A. polytricha* and other wild and cultivated edible mushrooms were reported to have higher levels of K and P than *A. thailandica*. *Auricularia thailandica* has also been shown to accumulate similar amounts of Mg and Na as other wild and cultivated edible mushrooms (Kalač 2010, 2013). However, the Ca content of *A. thailandica* is higher than the 100–500 mg/kg levels which have been found in other wild and cultivated edible mushrooms (Kalač 2010, 2013). High concentrations of Fe and Zn have been found in *A. thailandica*, while the presence of Cr, Cu, Mn and Ni has been reported in less than 20 mg/kg of dry weight (Table 3). Similar observations have been reported for *A. auricula-judae* (Kadnikova et al. 2015) and *A. polytricha* (USDA 2015). Macro- and microelements are essential to both fungi and consumers of fungi; however, the accumulation of toxic elements adversely affects these consumers, and humans in particular (Wang et al. 2014). Various concentrations of toxic metals have been reported in *Auricularia* spp. from different countries (Fang et al. 2014; Latiff et al. 1996; Okwulehie and Ogoke 2013). Quantities of toxic elements in *A. thailandica* (Table 3) were observed to be within safe limits according to the levels prescribed for foodstuffs by European Union Commission (EC) regulations (Table 7) (European Commission 2006). According to provisional tolerable weekly intake levels produced by the FAO/WHO (Kalač 2010), *A. thailandica* is a safe food for both children and adults (Table 7). However, toxic metal accumulation by mushrooms depends on species, substrate composition and environmental factors (Cheung 2010; Falandysz and Borovička 2013). The substrate composition we used to grow *A. thailandica* was responsible for having safe levels of toxic metals in the mushrooms.

The reducing power of aqueous and methanol extracts depends on the concentration of the extract. We found that methanol extract of *A. thailandica* had higher reducing power than aqueous extract. The reducing power of a compound may serve as a significant indicator of its potential antioxidant activity (Meir et al. 1995). Methanol extract of *A. thailandica* exhibited significantly higher free radical scavenging effects compared to the aqueous extract. The scavenging effects increased within a certain range of concentrations: free radicals were efficiently scavenged in concentrations of between 1–3 mg/mL in both methanol and aqueous extracts of *A. thailandica* (Fig. 4). It was reported that the methanol extracts of cultivated *A. fuscosuccinea* (white variety) and *A. polytricha* are effective at DPPH radical scavenging (Lin et al. 2013). *Auricularia polytricha* was more effective than *A. thailandica*, which was more effective than *A. fuscosuccinea* (white variety). Free radicals are formed in

Table 6 Nutrient content of cultivated *Auricularia* spp. (% of dry weight), as shown in previous studies investigating the nutrition of *Auricularia* spp.

Nutrient constituent	*A. auricula-judae*	*A. fuscosuccinea*	*A. polyticha*	*A. thailandica* (this study)	References
Fat	1.5, 1.7	4.5 (b, w)	0.48, 0.5	2.93	Crisan and Sands (1978); Kadnikova et al. (2015); Mau et al. (1998); USDA (2015)
Fibre	3.5, 6.9	11.7 (b), 12.5 (w)	3.6	4.62	Aletor (1995); Crisan and Sands (1978); Mau et al. (1998)
Protein	12.5, 11.5	8.6 (b), 12.5 (w)	6.5	12.99	Crisan and Sands (1978); Kadnikova et al. (2015); Mau et al. (1998); USDA (2015)
Free amino acids	–	0.12 (b), 0.07 (w)	0.05	1.13	Mau et al. (1998)
Total soluble sugars	10.2	9.9 (b), 10.9 (w)	17.9	16.23	Kadnikova et al. (2015); Mau et al. (1998)

b Brown strain, *w* white strain

Table 7 The comparison of toxic elements recorded in *Auricularia thailandica* with tolerable and maximum intake levels for certain toxic elements prescribed by the Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO) and the European Union Commission (EC) regulation no. 1881/2006

Toxic element	Maximum level mentioned in EC (mg/kg)	Provisional tolerable weekly intake FAO/WHO (mg/kg body weight)	Quantities of toxic elements recorded in *A. thailandica* (this study) (mg/kg)
As	0.1	0.007	0.0690 ± 0.0037
Cd	0.2	0.025	0.0672 ± 0.0012
Hg	0.5	0.015	0.0005 ± 0
Pb	0.3	0.005	0.0536 ± 0.0041

cellular oxidation reactions and are associated with lipid and protein peroxidation, resulting in cell structural damage, tissue injury or gene mutation, leading to the development of various degenerative diseases, such as cancer, cardiovascular disease, immune system decline, brain dysfunction, cataracts and ageing (Asatiani et al. 2010). Due to the effective free radical scavenging activity of ABTS, DPPH and its reducing power, *A. thailandica* can be considered a naturally safe antioxidant.

Phenolic compounds exhibit antioxidant activity in vitro and in vivo (De Silva et al. 2013). It has been reported that phenolic compounds are responsible for the antioxidant activities of mushrooms (Palacios et al. 2011). Our results showed that the total phenolic content of methanol extract of *A. thailandica* is significantly higher than aqueous extract of cultivated *A. thailandica* (Table 4). The total phenolic content in methanol extract of *A. fuscosuccinea* (white variety) and *A. polytricha* were 10.8 mgGAE/g and 2.98 mgGAE/g, respectively (Lin et al. 2013), which were both higher than *A. thailandica* (1.99 ± 0.01 μgGAE/mg). Although phenolic compounds are responsible for the antioxidant activity of mushrooms, the inhibition extent does not correlate with the total phenolic content (Palacios et al. 2011). This is because polysaccharides of *A. auricula* also exhibit considerable antioxidant activity (Zhang et al. 2011).

Auricularia thailandica has several advantages for consumers over other commercially available jelly mushrooms, including higher nutrient content especially rich in proteins and essential amino acids, higher antioxidant levels and larger fruiting bodies. Research on the domestication of *A. thailandica* using cheaply available agricultural waste could provide new insights to the mushroom growing industry. However, further studies are required to identify the mix of agricultural wastes which can result in optimal biological efficiency. A large number of wild edible mushroom species have been recorded in Southeast Asia (Thawthong et al. 2014), and it is likely that there is great potential in this region for further domestication research, which could help fulfill future demand for edible mushrooms.

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